# STUDY GUIDE FOR MATH 195 MIDTERM 2

## Format and formula sheet

The exam has 14 questions. Most questions are calculational, but expect to see a few which ask you to know important definitions.

No notes nor electronic devices are allowed. However, as part of the exam you will be given a formula sheet. It will contain four of the following formulas. As a class during review day you will choose which four you want.

(1) The definitions of secant, cosecant, and cotangent.
(2) The law of sines.
(3) The law of cosines.
(4) The even-odd identities.
(5) The variants of the pythagorean identity for tangent/secant and cotangent/cosecant.
(6) The angle-sum identities.
(7) The double-angle identities.
(8) The half-angle identities.

## Topics to know

You should know everything we've learned about trigonometry. This includes:

- How to work with angles and the unit circle.
- The unit circle definition of the trig functions.
- The right triangle definition of the trig functions.
- How to use the laws of sines and cosines to solve triangles.
- How to use trig functions to model waves. This includes graphing waves, coming up with formulas based on a graph or verbal description of a wave, and getting big picture information (increasing, etc.) from a graph of a wave.
- How to solve equations involving trig functions.
- How to use trig identities to rewrite and simplify formulas.
- How to use the angle-sum and half-angle identities to calculate new values of trig functions.

You should know the values of trig functions at the cardinal directions, but you are not being asked to memorize others. Either you can leave answers in exact form (e.g. $\cos(\pi/4)$) or I will provide the values.

## Sample questions

(1) What quadrant is the angle $17\pi/7$ in? What is its reference angle?
(2) Determine two different angles coterminal with the reference angle to $-4\pi/5$.
(3) You know that an angle $\alpha$ is in quadrant II and $\cos(\alpha) = -3/5$. Determine the values of the other five trig functions at $\alpha$.
(4) You know that an angle $\beta$ is in quadrant III and $\cot(\beta) = 1/4$. Determine the values of the other five trig functions at $\beta$.
(5) Knowing that $\sin(\pi/6) = 1/2$, determine $\sin(5\pi/6)$, $\sin(7\pi/6)$, and $\sin(11\pi/6)$.
(6) . A right triangle with angle $\alpha$ has the side opposite the angle with length 5 and a hypotenuse of length 13. Determine the value of all six trig functions at $\alpha$.
(7) A plane travels straight east for 200 miles. Then it turns to travel southeast for 100 miles. How far is it from its start place? Simplify your answer as much as possible.
(8) A triangle has a side of length 4 in between angles of $30°$. Determine the missing angle and side lengths. Simplify your answers as much as possible.
(9) A triangle has sides of length $2, 3, 4$. Determine its angles. Simplify your answers as much as possible.

(10) You know the two legs of a right triangle are $a$ and $b$. What does the law of cosines tell you is the hypotenuse? Fully simplify your answer.

(11) You know an angle $\alpha$ and hypotenuse $c$ of a right triangle. What does the law of sines tell you is length of the leg opposite $\alpha$? Fully simplify your answer.

(12) Sketch a graph of the following trig function, showing one full period.

$$a(t) = -3\tan(2t)$$

(13) For the following wave, determine its amplitude, centerline, and period. Sketch a graph, showing one full period.

$$b(t) = 2\sin(\pi t) - 1$$

Using this graph, determine on the period you graphed where $b(t)$ is increasing, decreasing, concave up, and concave down. Answer in interval notation.

(14) For the following wave, determine its amplitude, centerline, and period. Sketch a graph, showing one full period.

$$c(t) = -\cos(t/4) + 1$$

(15) For the following wave, determine its amplitude, centerline, and period. Sketch a graph, showing one full period.

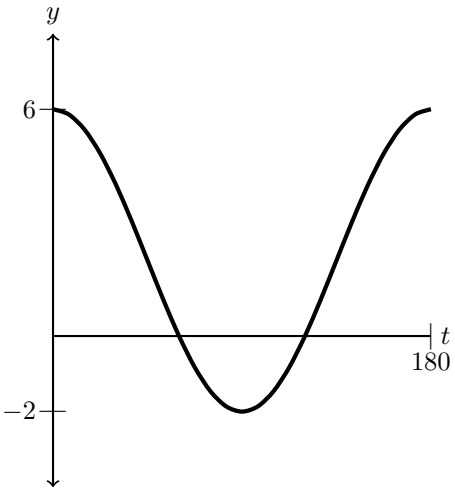
$$d(t) = -3\sin(4t) - 2$$

On this period, where is $d(t)$ increasing? Decreasing? Concave up? Concave down?

(16) A wave has centerline $y = 1$, amplitude 3, and period 4. If at time $t = 0$ the wave is at its lowest point, write a function modeling the wave.

(17) A wave has minimum $y = 1$, maximum $y = 11$, and period $3\pi$. If at time $t = 0$ the wave is at the midpoint moving downward, write a function modeling the wave.

(18) The graph of one period of a wave is shown below. Write a function modeling the wave.




(19) Find the general solution to the equation

$$\sin t = 0.$$

(20) Find the general solution to the equation

$$\cos t = 0.$$

(21) Find the general solution to the equation

$$\tan t = 0.$$

(22) Find the general solution to the equation

$$2\sin(t/3) - 3 = 4.$$

[Hint: $\sin(\pi/6) = 1/2$.]

(23) Find the general solution to the equation

$$3\cos(2t) = 4.$$

(24) Find the general solution to the equation

$$2\tan(3t - \pi/2) + 1 = 4.$$

(25) A wave is described by the following function. Find all times $t$ where $y(t) = 2$.

$$y(t) = 4\cos(\pi t) - 1$$

(26) Simplify the following expression as much as possible.

$$\frac{\sin x \cos x}{\tan x} - 1$$

(27) Simplify the following expression as much as possible.

$$\frac{\sin t \cot t \cos t + 1}{\tan t}$$

(28) Simplify the following expression as much as possible.

$$\cos x - \cos^3 x$$

(29) Simplify the following expression as much as possible.

$$\tan x \sin x + \sec x \cos^2 x$$

(30) Determine $\tan(\pi/8)$, knowing the values for the trig functions at $\pi/4$.

(31) Explain why

$$\cot(2x) = \frac{\cot x}{2} - \frac{\tan x}{2}.$$

(32) Using the other identities explain why

$$\sin(x/2) = \pm\sqrt{\frac{1 - \cos x}{2}}.$$